

Arnob Banik

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PROFESSIONAL SUMMARY

Technical professional with experience spanning **enterprise IT, full-stack software development, and applied AI**. Built and deployed production applications and workflow systems using Next.js, TypeScript, PostgreSQL, Supabase/Neon, REST APIs, authentication, scheduled serverless functions, and Gemini/OpenAI integrations. At Banglalink, supported **Oracle Fusion ERP Cloud** Financials and Procurement, system operations, and ISO 27001 audit readiness, including consolidating approximately **15,000 tax conditions to 460**. Currently completing an MSc in Computer Science (AI), with research focused on efficient multimodal retrieval using CLIP, HNSW, and color-aware prefiltering.

EXPERIENCE

- Banglalink** Dhaka, Bangladesh
Enterprise Solutions and Services Specialist Engineer, IT Sept 2023 – June 2024
 - Consolidated approximately 15,000 Oracle ERP tax conditions into 460 maintainable rules, improving time-to-market and reducing calculation issues.
 - Supported Oracle Fusion ERP Cloud Financials (AP, AR, GL) and Procurement workflows, resolving operational issues and maintaining vendor/supplier and financial documentation.
 - Prepared ISO 27001 audit artifacts across risk management, incident management, access control, and backup documentation, contributing to a 90% audit score.
- Banglalink** Dhaka, Bangladesh
Information Technology Intern June 2023 – Sept 2023
 - Developed an internal IT-support chatbot and assisted with Oracle ERP financial/vendor documentation, EDMS configuration, roles, and access permissions.
- GAOTek Inc.** Remote
Software Development Intern – Team Leader Dec 2022 – March 2023
 - Led a remote intern team delivering Angular components; triaged defects, reviewed code, and supported onboarding and task coordination.

SKILLS

- Languages:** Python, TypeScript, JavaScript, SQL, R, MATLAB
- Full-Stack & APIs:** Next.js, React, Node.js, REST APIs, Tailwind CSS, Angular, Auth.js, Zod, Prisma ORM, PWA, Capacitor
- Data & Backend:** PostgreSQL, Supabase, Neon, Row-Level Security (RLS), Edge Functions, pg_cron
- Applied AI & ML:** Gemini/OpenAI APIs, Machine Learning, Deep Learning, BERT, CLIP, HNSW, FAISS, scikit-learn
- Cloud, DevOps & Enterprise:** Vercel, OCI, GitHub Actions/CI, Postman, Oracle Fusion ERP Cloud (AP, AR, GL, Procurement), BI Publisher, Visual Builder, JIRA, Confluence, ISO 27001

PROJECTS

- Job Application Intelligence Dashboard** / *Next.js, TypeScript, Supabase/Postgres, Gemini/OpenAI, Gmail, Vercel* / *GitHub*
 - Engineered a human-in-the-loop job intelligence system that collects public postings from ATS feeds and employer career sites, applies deterministic eligibility filters and profile-based ranking, tracks opportunities through an application funnel, and maintains a 148-company target-employer watchlist.
 - Implemented Supabase/PostgreSQL persistence with RLS and four scheduled Edge Function workers via pg_cron/Vault; added Gemini-first analysis with optional OpenAI, ATS-friendly PDF generation, Gmail digests/draft outreach, and privacy/claims safeguards.
- Flowdesk – Full-Stack Family CRM** / *Next.js, React, TypeScript, Prisma, Neon, Auth.js, Capacitor* / *Live*
 - Designed and shipped Flowdesk as a production household CRM, installable PWA, and Android app with 40+ authenticated API routes, household-scoped multi-user data isolation, invite onboarding, and Google OAuth.
 - Built Kanban task management, calendar, finance/budgets, medication schedules and dose logs, notes, contacts/projects, focus tools, and Web Push/email alerts; deployed with Prisma, Neon Postgres, and Vercel.
- MSc Thesis – Color-Aware Composed Image Retrieval** / *Python, CLIP RN50x4, HNSW, FashionIQ*
 - Developed a color-aware composed image retrieval framework using fine-tuned CLIP RN50x4, HNSW, and voxelized RGB prefiltering on FashionIQ; evaluated prefilter/postfilter designs using computational-overhead and retrieval-efficiency measures.
- Phishing URL Detection Using Artificial Intelligence** / *Python, ML, BERT, Flask*
 - Trained ML and BERT-family models on approximately 235,000 URLs, achieving 99%+ precision and recall; deployed a Flask interface for interactive prediction.

EDUCATION

- University of Windsor** Windsor, Ontario, Canada
Master of Science in Computer Science (Artificial Intelligence Specialization) Sept 2024 – Aug 2026 (Expected)
- Vellore Institute of Technology** Vellore, Tamil Nadu, India
Bachelor of Technology – Computer Science and Engineering; GPA: 8.20/10 July 2019 – July 2023

CERTIFICATIONS

- Google IT Support
- Oracle Cloud Infrastructure 2023 Certified Data Science Professional
- Oracle Cloud Infrastructure 2023 Certified Foundations Associate
- Oracle Cloud Data Management 2023 Certified Foundations Associate
- AWS Academy Graduate – AWS Academy Cloud Foundations
- Machine Learning for All